CRITICAL THINKING IN THE AGE OF AI

An Integrated Symposium
Prospectus

Carl Raschke, University of Denver | Jordan Loewen-Colón, Queens University | Adam D.J. Brett, Syracuse University

EXECUTIVE SUMMARY

American higher education hovers at an epochal inflection point. Generative artificial intelligence has broken the foundational warrant on which the university built its authority — the assumption that an academic artifact certifies the capacity that produced it. When a machine can generate the essay, the problem set, or the research summary, the artifact no longer proves what it claimed to prove. The credential drifts free of the competency it was meant to signal.

One capacity survives and, in surviving, becomes more important than ever — the ability to think critically. Yet the expression "critical thinking" is more frequently invoked than elaborated, more sloganized than captured in the concrete and quotidian production of knowledge — and the constitution of the social knowledge base. A recent intervention in *Inside Higher Ed* goes so far as to argue that higher education has never genuinely taught critical thinking at all.[1] The present symposium takes that provocation seriously and proposes to do something about it.

The Whitestone Foundation through its "Knowledge Futures" initiative" proposes to convene a working symposium entitled "Critical Thinking in the Age of AI". The symposium will bring together philosophers of mind and technology, cognitive scientists, pedagogical theorists, and practitioners across post-secondary sectors to address a single integrating question — what does critical thinking require, and how must it be taught, when artificial intelligence mediates so much of cognition and knowledge generation?

The online gathering is designed as an ongoing a working session. Its deliverables are a prioritized agenda of the ten most important questions the new state of knowledge production insinuates, a curated resource pack, a white paper synthesizing the proceedings, and the seed of a curriculum that can be adapted across post-secondary contexts. It is, simultaneously, the first proof of concept for the Knowledge Futures platform the Whitestone Foundation is assembling.

THE THEORETICAL CHALLENGE

A. The Warrant Crisis

The university has always rested on what may be called the warrant-by-artifact principle. A degree certifies competency because it certifies work done. The essay proves that the student did the work. The examination proves that the student knows. The dissertation proves that the student can produce original knowledge. These claims could

[1] Natalia Pasternak, "We Have Never Taught Critical Thinking." *Inside Higher Ed*, June 23, 2026. https://www.insidehighered.com/opinion/views/2026/06/23/we-have-never-taught-critical-thinking-opinion

be taken seriously because the artifact and the capacity were inseparable. You could not produce the one without exercising the other.

Generative AI severs this connection. The artifact arrives without the implicit exercise. The essay is produced, but the thinking process that the composition of essay was intended to activate and demonstrate does not follow. The credential, therefore, no longer proves what it purports to certify. Such a default of critical intelligence is the operative machinery of American higher education as of this writing across almost every school, sector, and discipline.

> *AI severs the link between the artifact and the capacity. The response is not to ban the artifact but to relocate the warrant: to migrate it from what the student produces to how the student thinks.*

The response that many institutions have reached for first — detection, prohibition, honor-code enforcement — treats the symptom. The question the symposium addresses is endemic. Given that AI is not going away, what becomes of the institution whose authority depended on the inseparability of artifact and the capacity to create it? And what must critical thinking mean once that inseparability cannot be presupposed any longer?

B. Critical Thinking as the Durable Capacity to Innovate

The guiding thesis behind this symposium is that critical thinking is precisely the capacity that AI cannot replicate and that its practice therefore cannot be offloaded without cost. The empirical case is building for such a claim. Research at the MIT Media Lab found that participants who relied on AI for essay writing showed the weakest neural connectivity and the lowest sense of ownership of their own work, and that those who began with AI struggled to recover independent performance when switched to unaided conditions. What the researchers called cognitive debt is the atrophy that follows from outsourcing judgment.[2]

The philosophical case runs deeper. Shannon Vallor's concept of moral and intellectual deskilling names the erosion of the capacity for judgment when deliberation is increasingly outsourced to systems that are not capable of practical wisdom. In her framing AI systems are mirrors. They reflect back the recorded patterns of human thought and judgment, but they cannot originate. The danger is not that the mirror lies, though it does on occasion, but that we mistake the reflection for the persona — i.e., the labor of thinking itself and lose the habit of doing the latter.

What critical thinking is, however, is less clear than these assessment might suggest. It is defined by employers and accreditors, postured through syllabi, and appraised by

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[2] Nataliya Kosmyna, et al. "Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task." *arXiv,* June 2025.

rubrics, yet seldom theorized with any genuine nuance or precision. Rebekka King's account of critique as a context-dependent competency — one that evolves through practice, necessitates a specific stance toward one's own governing assumptions, and cannot be reduced to a transferable algorithm — is closer to what the moment demands than most institutional formulations. The symposium will treat the definition as an open question rather than a settled premise.

C. The Omnibus Challenge to Higher Education

The theoretical challenge AI poses to American higher education is not sector-adjacent. It applies to research universities, liberal arts colleges, regional universities, community colleges, and professional schools. Each faces a distinct variant of the same problem.

The research university faces the most direct assault on its knowledge-production warrant. If AI can generate plausible and coherent research summaries, literature reviews, and even preliminary analyses, what distinguishes the trained researcher from the machine-assisted neophyte? The liberal arts college faces the most profound challenge to AI's core claim, namely, that the artifact of education is not the degree but the whole person, shaped by the discipline of reading, the competency of writing, and the aptitude for persuasive argument. The community college faces the most urgent practical demand. It students are often already in the workforce, are in need of the wherewithal to use AI tools impactfully and skillfully, and cannot afford a leisurely "rethink" of the meaning and purpose of their lives, let alone the curriculum.

The comprehensive regional university faces all three versions simultaneously, insofar as it serves students with vocational aims, research aspirations, and general education requirements under a single institutional umbrella. Professional schools are confronted with the question of what becomes of curated expertise, such as in law, medicine, business, engineering, when AI can perform the procedural and even analytical work that the professional claimed as exclusive expertise.

No single curriculum model fits all these contexts. The symposium is therefore designed to produce not a single answer but an adaptable framework of queries, resources, and overarching procedures and principles that each sector might adapt. The aim is an architecture for the synergy of critical thinking and AI training that is not sector dependent (rather it is "sector agnostic") in its theoretical marrow and translatable into sector-specific curricula in its actual deployment.

D. AI as Both Subject Matter and Instrumentality

The symposium's framing distinguishes three postures toward AI that any serious curriculum involving critical thinking must cultivate. Students and practitioners must learn to think critically about AI — to understand what these systems do, how they work, what they cannot do, and what are the stakes in relying on them. They must learn to think with AI — to use these tools in ways that amplify rather than substitute for their own judgment, treating AI as a "co-intelligence" rather than an oracle. And they must learn to think despite AI — to maintain the autonomous habits of cognition that atrophy when outsourcing becomes the easy fallback.

These three postures are not alternatives to each other. They are concurrent approaches that a critical thinker in an AI-saturated environment must hold in tension. The curriculum the symposium seeds will be organized within this tripartite scaffolding.

The Wild Globalization project initiated by Gary Bedford furnishes the conceptual matrix. AI is not merely a classroom tool. It is a transformative force in the global knowledge economy, redistributing cognitive labor, reshaping what counts as expertise, and challenging the institutions that have historically shaped the growth of cognitive capital. Critical thinking in the age of AI is therefore not merely another tweak to the norms of classroom instruction and pedagogical advancement. It is a systemic response to a civilizational shift.

GOALS AND OBJECTIVES

Primary Goals

- Convene a working symposium that produces a shared, prioritized agenda rather than a collection of individual opinions.
- Define critical thinking with precision adequate to the AI moment, distinguishing it from procedural compliance, information retrieval, and pattern matching.
- Develop a sector-agnostic curriculum framework for teaching critical thinking in post-secondary settings that both utilizes and trains in AI, translatable across research universities, liberal arts colleges, comprehensive universities, community colleges, and professional schools.
- Address the theoretical challenge of integrating AI into the current architecture of American higher education without privileging any single sector.
- Generate durable public artifacts — a white paper and, on a longer horizon, a book — that advance the conversation beyond the symposium space.
- Launch a knowledge-futures platform that can host curricula, convene future gatherings, and build a continuing community of practice.

Subsidiary Objectives

1. Identify and rank the ten most important questions in critical thinking and AI, each with a rationale and a resource pack.
2. Establish a working definition of critical thinking adequate to the AI context.
3. Map how the challenge differs and how it is the same across post-secondary sectors.

4. Propose specific pedagogical practices for each of the three postures — thinking about, with, and despite AI.
5. Identify the institutional factors — curriculum, assessment, policy — that support or obstruct critical thinking in AI-integrated environments.
6. Assign ownership of each top ten question and schedule follow-up.

SYMPOSIUM METHOD AND AGENDA

A. Selection Principles

The value of this symposium lies in the strategic mix of its participants. The gathering must hold philosophers and humanists, cognitive scientists, pedagogical theorists, and working practitioners in the same room, in roughly equal standing. It must represent post-secondary education across sectors without privileging any particular one. And it must be small enough for functioning conversation — a target of ten to twelve participants — while broad enough to steer clear of lazy-minded consensus and a de facto groupthink.

Three specific principles guided participant selection. First, the subject of each participant's contribution must be critical thinking or human judgment, not AI per se as a principal subject. This consideration prevents the symposium from drifting into a panel of tech nerds. Second, participants must be accessible, either through existing warm relationships or through academic-tier honoraria,consistent with a first-year venture operating without corporate keynote budgets. Third, the roster must include at least one voice skeptical that AI changes the critical-thinking question fundamentally, as a check against assuming the conclusion.

B. Format

The symposium runs as a structured single working session of four to five hours, designed to generate ideas widely and then to converge with discipline. It is not a panel. No participant gives a prepared talk. Participation is a condition of attendance. A facilitator holds the process; Carl Raschke opens and closes with framing remarks. The session can be held in person, virtually, or in a hybrid format. An initial virtual session is recommended to reduce costs while building community before a subsequent in-person gathering.

C. Pre-Planning

Distributed two weeks before the session:

- Three competing definitions of critical thinking drawn from cognitive science, philosophy, and pedagogical theory.
- The Gerlich (2025) paper on cognitive offloading and critical thinking.
- The MIT Media Lab summary of the Kosmyna et al. (2025) cognitive-debt study.

- The Inside Higher Ed provocation piece on whether higher education has ever taught critical thinking.
- Two preparation questions: (1) What is the most important thing AI changes about how critical thinking must be understood? (2) What is one thing it does not change?

Each participant is asked to arrive with one live question and one resource they believe the group has not seen.

D. Ground Rules

- Charitable disagreement: challenge the idea, not the person.
- Intellectual humility: state confidence levels; revise in public.
- Deferred judgment during ideation: no evaluation while generating.
- Every voice on the board: the facilitator ensures no participant dominates.
- The three postures as organizing lenses throughout: thinking about, with, and despite AI.

VI. Curriculum Framework

A. The Three-Posture Model

The curriculum framework organizing principle is the three postures introduced in the theoretical argument. Any post-secondary critical-thinking curriculum adequate to the AI moment must develop all three simultaneously, not as a sequence.

Thinking About AI

Students must understand what these systems are — and are not. This includes 1) how large language models work and what they do statistically rather than cognitively 2) where they are reliable and where they hallucinate; how they are trained and on whose data 3) what the governance and democratic stakes are of their proliferation. This posture is the analytical and evaluative dimension of critical thinking applied to the tool itself. It is the subject matter of Shannon Vallor's moral deskilling argument and Gary Marcus's technical-limits critique. You cannot use a tool critically if you do not understand what it is doing.

Thinking With AI

Students must develop the practical capacity to use AI as a genuine thinking partner rather than as an answer machine. This means learning to prompt with precision, to evaluate AI output critically rather than accepting it passively, to iterate in dialogue with the system, and to bring their own judgment to bear on what the AI produces. Ethan Mollick's co-intelligence framing is the relevant pedagogical model here — AI as co-worker and coach, not oracle. The key pedagogical shift is from evaluating the artifact the AI produces to evaluating the student's own reasoning about and with that artifact.

Thinking Despite AI

Students must maintain and develop the independent cognitive habits that atrophy under sustained AI reliance. Nataliya Kosmyna's cognitive-debt finding is the empirical starting point. The capacity weakens when not exercised. Curriculum in this posture requires deliberate unassisted practice — writing, argument, analysis, and problem-solving done without AI support — not as a prohibition but as cognitive exercise, the way an athlete trains without the equipment to develop the underlying strength. Rebekka King's account of critique as a context-dependent competency built through practice is the theoretical grounding.

B. Sector Translation

The three-posture framework is sector-agnostic at its theoretical core. Its translation into curriculum is sector-specific.

Research Universities

The critical-thinking challenge at research universities centers on the knowledge-production warrant — what distinguishes original research from AI-assisted synthesis? Curriculum in this context should emphasize the distinction between pattern recognition and authentic inquiry, the epistemology of source evaluation and citational responsibility, and the specific thinking practices — e.g., hypothesis formation, experimental design, interpretive argument — that AI cannot substitute for. Assessment must migrate from the artifact (the paper, the lit review) to the process (the research journal, the oral defense, the iterative revision documented over time).

Liberal Arts Colleges

The liberal arts context is where the claim that education produces a person, not a credential, is most expressly made and most immediately threatened. Curriculum here should foreground the argument that reading, writing, and argument cultivate a certain kind of person, and that AI-assisted dodges short-circuit that formation. The critical-thinking posture of thinking despite AI is especially salient here. Students must experience the difficulty of unaided intellectual work as constitutive of education, not merely as a test of honesty. Seminar discussion, oral examination, and portfolio assessment that documents process over time are the appropriate assessment forms.

Comprehensive Regional Universities

Comprehensive universities serve the broadest range of students and the widest spectrum of vocational and academic aims. Curriculum here must be maximally translatable — modular, scalable, and adaptable across disciplines. The three-posture framework fits this context well because it does not require a dedicated course listing. It can be embedded in existing discipline-specific courses as a metacognitive overlay. Faculty development is the leverage point. Training instructors across disciplines to teach critical thinking about, with, and despite AI within their own subject matter.

Community Colleges

Community colleges face the most urgent and the most under-resourced version of this challenge. Their students are often already in the workforce, pressed for time, and most directly exposed to AI tools in the jobs they hold and seek. Curriculum at this level must be practical without being merely procedural: it must give students genuine critical-thinking tools, not just AI-use tips. The thinking with AI posture is the most immediately useful point of entry. Thinking about AI provides the evaluative framework, while thinking despite AI provides the cognitive disciplinary ballast for AI applications. Stackable credentials and short-form modular curricula are the appropriate delivery formats.

Professional Schools

Professional schools — law, medicine, business, engineering, social work — face the erosion of professional judgment as AI performs increasingly sophisticated procedural and analytical work. Curriculum here must address what professional expertise actually is once the procedural work is automated, and it must do so at the level of the discipline's specific reasoning practices. Legal reasoning, clinical judgment, financial analysis, and engineering problem-solving each have distinctive critical-thinking structures that AI challenges in specific ways. Simulation, case study, and supervised practice remain the appropriate pedagogical forms; the critical-thinking curriculum must be embedded in them rather than standing apart.

C. Assessment Principles

Across all sectors, the assessment shift the curriculum framework requires is the same in kind: from artifact to process, from product to documented thinking. Specific instruments include:

- Process portfolios that document the student's thinking over the duration of a course or project, including drafts, AI transcripts submitted alongside human revisions, and reflective commentary on how the student's thinking developed.
- AI-transcript submission as a required component of AI-assisted work, paired with a student account of what the student accepted, rejected, and modified and why.
- Oral examination and defense as the primary site of warrant for research and argument, since the student must demonstrate, in real time and without AI support, that the thinking is genuinely theirs.
- Unassisted in-class writing and problem-solving maintained as a regular cognitive-discipline practice, framed not as a prohibition but as the training equivalent of the athlete's unaided workout.
- Collaborative peer critique that requires students to evaluate each other's reasoning publicly, developing the evaluative posture that is the core of critical thinking.

DELIVERABLES

Near-term (within one month of symposium)

- Top ten questions — synthesized list with rationale and resource pack, owner assigned to each item.
- Working definition — a provisional definition of critical thinking adequate to the AI context, agreed by the group.
- Cross-sector curriculum principles — the sector-agnostic framework in summary form.

Short-term (within six months)

- White paper — the primary public-facing synthesis of the symposium. Audience: higher-education administrators, faculty, and policy-makers. Tone: serious, accessible, non-technical. Author: the group collectively, drafted by Carl Raschke and Adam DJ Brett.
- Curriculum module prototype — a first version of the three-posture curriculum module in a format translatable across sectors, suitable for faculty development use.

Long-term (within eighteen months)

- Book — a longer work developing the white paper arguments, incorporating participant essays and the curriculum framework in full. Primary long-term reputation-building artifact.
- Annual Symposium — the first gathering recurs, with a rotating question or sector focus, building the community of practice and the Whitestone Foundation's convening reputation.
- Online Curriculum Platform — hosted through the Whitestone LXP, making the curriculum framework available in modular, adaptable form to institutions and individuals.

LOGISTICS AND BUDGET PRINCIPLES

Format recommendation

The first symposium should be conducted virtually to minimize costs and maximize participant accessibility. A half-day video session (four to five hours) with pre-work distributed two weeks prior is the recommended format. An in-person follow-up gathering — whether a second symposium or a working session tied to an existing conference such as the AAR annual meeting — should follow within six to twelve months. Virtual format also permits recording and subsequent public release of edited proceedings, which extends the public reach of the deliverables.

Funding Model

The symposium is funded through private donors invested in making a difference in higher education, consistent with the Whitestone Foundation's nonprofit model. Prospective supporters should be approached with a prospectus describing the initiative as already in motion (which it is) and invited to participate in shaping its direction. Grant funding from foundations with higher-education or civic-intelligence mandates is a secondary avenue worth exploring for subsequent gatherings.

PLACEMENT WITHIN THE KNOWELDGE FUTURES INITIATIVE

This symposium is not a standalone event. It instantiates in a single project nearly every pillar of the Whitestone Foundation's Knowledge Futures platform.

- Publishing platform — the white paper and the book are the first major publications under the Knowledge Futures banner.
- Recurring symposium — the first gathering seeds an annual convening that builds reputation and community over time.
- Academy and curriculum — the three-posture framework and sector-specific modules are the first substantive curriculum offerings on the LXP.
- Scholars in residence — symposium participants become the first cohort of affiliated scholars, building the network from which a more formal fellows program can grow.

Substantively, this is the higher-education strand of the broader Wild Globalization agenda. It treats the university not as a subject to defend but as a case study in how knowledge institutions must reconstitute themselves when their old warrants fail. What the group learns here informs the wealth, ecology, and international security strands that sit alongside it under the Knowledge Futures umbrella.

The ultimate claim of the entire initiative, articulated most plainly, is as follows: Whitestone is teaching people how to think, in a world where machines have made thinking both more important and more endangered than at any previous moment in the history of education. That is the sentence on which donor conversations, symposium invitations, and curriculum design all converge.

X. Next Steps

- Finalize the participant list and issue personal invitations, leading with existing warm relationships.
- Confirm honoraria budget and identify a funding source for the first gathering.
- Identify the unnamed slots: cognitive scientist / philosopher of mind; community college practitioner.
- Prepare and distribute the pre-work package two weeks before the session.

- Confirm facilitator and logistics (platform, date, recording permissions).
- Convene the symposium.
- Assign the white paper lead and resource-pack owner at the close; set follow-up date.

- Present the curriculum module prototype to the Thursday Club for review within three months.
- Approach prospective donors with the prospectus alongside this plan.